NETWORKWALKS ACADEMY • TECHNICAL INTERNSHIP DELIVERABLE

CYBERSECURITY INTERNSHIP

Cohort B084 • Week 2 Milestone Project

Subject: Information Gathering, OSINT & Network Footprinting

A structured technical assessment documenting practical passive and active reconnaissance exercises conducted against target domain `networkwalks.com` and private subnet `10.0.0.0/24`.

Candidate Identification & Project Metadata	
Candidate / Intern Name	**Dereje**
Program & Cohort	NetworkWalks Cybersecurity Internship (Cohort B084)
Module / Project Title	Week 2: Footprinting, DNS Enumeration & Perimeter Mapping
Primary Target (External)	`networkwalks.com` (IPv4: `192.232.216.135`)
Secondary Target (Internal)	`10.0.0.0/24` Lab Subnet (Nodes: `10.0.0.1`, `10.0.0.3`)
Assessment Toolkit	`whois`, `nslookup`, `whatweb`, `curl`, `wafw00f`, `dnsrecon`, `zenmap / nmap`
Submission Date	November 10, 2026
Deliverable Format	**Official Technical Write-Up (PDF Version 1.0)**

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1. Executive Summary

Information gathering constitutes the initial and most vital phase of the cybersecurity lifecycle, aligning with Phase 1 of the **Penetration Testing Execution Standard (PTES)** and the Reconnaissance stage of the **Lockheed Martin Cyber Kill Chain**. Thorough reconnaissance allows analysts and penetration testers to discover public-facing infrastructure, catalog technological dependencies, identify perimeter defense controls, and detect configuration exposures without triggering disruptive defensive counteractions.

During the Week 2 engagement of the NetworkWalks Cybersecurity Internship (Batch B084), hands-on assessments were carried out across two target environments:

- **External Web Target:** Investigated `networkwalks.com` using passive OSINT and active probing tools (`whois`, `nslookup`, `whatweb`, `curl`, `wafw00f`, and `dnsrecon`).
- **Internal Laboratory Subnet:** Surveyed the `10.0.0.0/24` private network using Zenmap (Nmap 7.991) to evaluate host responsiveness, port states, and operating system fingerprints.

Key Findings Summary:

- **Perimeter Defense:** The web target is actively defended by **ModSecurity (SpiderLabs)** Web Application Firewall (WAF).
- **Application Stack:** Built on WordPress (v7.1.4) and WordPress Download Manager (v3.3.76), served by Apache with an Nginx reverse caching proxy.
- **DNS Architecture:** Hosted on HostGator nameservers (`ns6135` / `ns6136.hostgator.com`). DNS queries revealed exact version disclosure: `BIND 9.16.23-RH`. DNSSEC is not implemented.
- **Security Headers:** Essential defensive HTTP headers (HSTS, CSP, X-Frame-Options) are absent.
- **Subnet Scan:** Active hosts `10.0.0.1` (identified as OpenBSD 6.9 PF firewall) and `10.0.0.3` enforce drop/filter firewall rules.

Target Asset	Network Location	Primary Technology	Defense Profile	Overall Posture
networkwalks.com	192.232.216.135	WordPress 7.1.4, Apache, Nginx Reverse Cache	ModSecurity WAF, HTTPS 301 Redirect	Moderate
mail.networkwalks.com	192.232.216.135	cPanel Mail Services, Autodiscover SRV	SPF Configured (SoftFail `~all`)	Moderate
ns6135 / ns6136.hostgator.com	50.87.144.87 / 192.232.216.131	BIND 9.16.23-RH	DNSSEC Unsigned, Version Disclosed	Needs Review
10.0.0.1	10.0.0.1 (Internal Gateway)	OpenBSD 6.9 (Packet Filter)	632 Closed, 368 Filtered Ports	Strong
10.0.0.3	10.0.0.3 (Lab Node)	Network Host / Appliance	1000 Filtered Ports (Silent Drop)	Strong

2. Reconnaissance Framework & Tool Taxonomy

The assessment strictly followed the reconnaissance phase specifications established by **NIST SP 800-115** (Technical Guide to Information Security Testing and Assessment) and the **Open Source Security Testing Methodology Manual (OSSTMM)**. Intelligence gathering was segmented into distinct passive and active operational disciplines:

Reconnaissance Category	Operational Methodology	Primary Objective	Tools Employed
Passive Reconnaissance (OSINT)	Interrogates publicly available third-party repositories, registries, and DNS resolvers. Generates zero direct traffic to target servers; undetectable by target IDS/IPS/WAF.	Identify domain ownership, registration timeline, IP blocks, and contact privacy.	`whois`, `nslookup` (via public Cloudflare resolver)
Active Web & DNS Footprinting	Directly sends HTTP/HTTPS requests and targeted DNS queries to target daemons to capture banners, HTTP response headers, and DNS zone records.	Fingerprint web servers, CMS frameworks, plugins, WAF engines, and DNS server software.	`whatweb`, `curl`, `wafw00f`, `dnsrecon`
Active Network Layer Scanning	Transmits crafted TCP SYN packets and ICMP echo probes across entire subnets to identify responsive nodes, evaluate port states, and profile TCP/IP stacks.	Discover live network appliances, identify firewall filtering policies, and fingerprint host OS kernels.	`zenmap / nmap`

Methodological Workflow:

1. **Passive Intelligence Gathering:** Query Whois and Recursive DNS for domain registrar, nameservers, and IP allocations.
2. **Perimeter Identification:** Run WAF fingerprinting (wafw00f) to detect filtering proxies before launching deep probes.
3. **Web Application Stack Profiling:** Interrogate HTTP response headers (curl) and application signatures (whatweb).
4. **DNS Infrastructure Enumeration:** Enumerate SOA, NS, MX, TXT (SPF), and SRV records; probe for BIND version strings.
5. **Network Layer Host & OS Mapping:** Scan internal subnet (Zenmap) using SYN stealth scans to determine firewall rules and OS kernels.

3. Detailed Technical Findings & Lab Execution

3.1 Task 1: Domain Registration & Ownership Footprinting (`whois`)

Objective: Perform passive OSINT footprinting against `networkwalks.com` to establish domain ownership metadata, registrar affiliation, creation/expiration milestones, authoritative name servers, and EPP anti-hijacking status codes.

```
$ whois networkwalks.com
```

Domain Name	`NETWORKWALKS.COM` (Registry Domain ID: `2452319255_DOMAIN_COM-VRSN`)
Sponsoring Registrar	GoDaddy.com, LLC (IANA Registrar ID: 146 | Abuse: `abuse@godaddy.com`)
Registration Timeline	Created: **2019-11-06** | Updated: **2025-11-12** | Registry Expiry: **2027-11-06**
Registrant Privacy Shield	Registration Private – Domains By Proxy, LLC (Tempe, Arizona, United States)
Authoritative Name Servers	`NS6135.HOSTGATOR.COM` and `NS6136.HOSTGATOR.COM`
EPP Domain Status Locks	`clientDeleteProhibited`, `clientRenewProhibited`, `clientTransferProhibited`, `clientUpdateProhibited`
DNSSEC Validation	`unsigned` (DNSSEC cryptographic signing is not configured at registrar level)

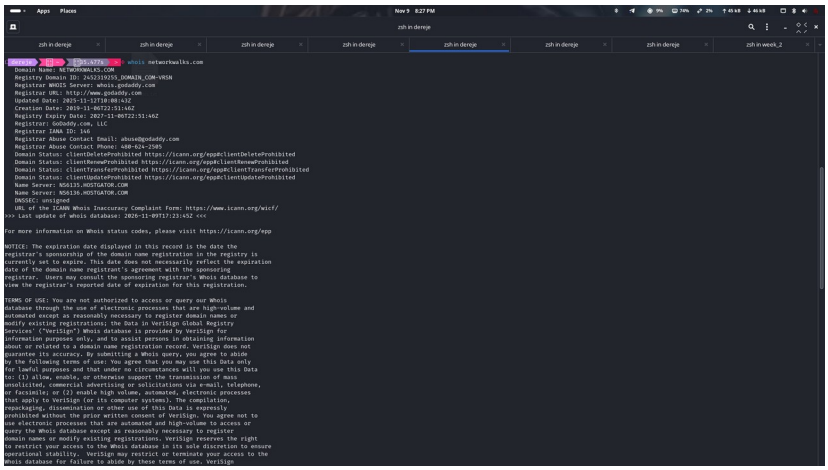


Figure 3.1: Terminal execution of whois networkwalks.com demonstrating domain registration and privacy protection.

Technical Security Analysis:

- **EPP Anti-Hijacking Protection:** The four client*Prohibited statuses indicate robust registrar locking. These controls prevent domain hijacking, unauthorized registrar transfers, and malicious DNS modifications.
- **Identity Protection:** Utilizing *Domains By Proxy, LLC* prevents malicious actors from harvesting administrative names, physical addresses, and direct phone numbers for spear-phishing or social engineering attacks.
- **Hosting Provider Infrastructure:** Delegation to HostGator nameservers establishes that the website is hosted on HostGator / Newfold Digital shared or dedicated server pools.

3.2 Task 2: DNS Resolution & Core Address Mapping (`nslookup`)

Objective: Resolve the domain's public IPv4 address using standard DNS querying through an independent recursive DNS resolver (Cloudflare 1.1.1.1) to analyze caching, latency, and hosting IP allocation.

```
$ nslookup networkwalks.com
```

```
Server:         1.1.1.1
Address:        1.1.1.1#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

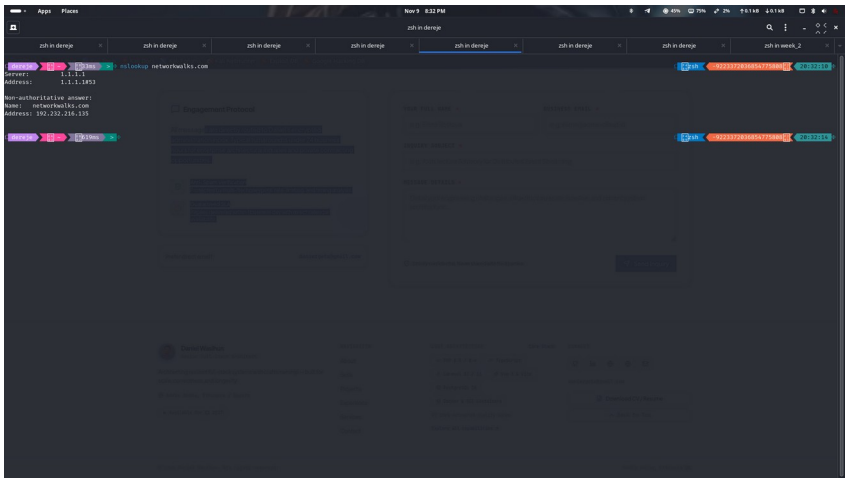


Figure 3.2: Terminal execution of nslookup resolving networkwalks.com to IPv4 address 192.232.216.135.

DNS Parameter	Observed Value	Technical Significance
Recursive Resolver	`1.1.1.1:53` (Cloudflare Public Anycast DNS)	Utilized an external non-authoritative resolver to prevent direct DNS logging on target authoritative nameservers.
Resolved IPv4 Address	`192.232.216.135`	Static public IPv4 address assigned to the target's primary web and mail ingress interface.
Autonomous System (ASN)	AS46606 – Unified Layer / HostGator	Confirms infrastructure hosting within HostGator's primary datacenter in Houston, Texas, USA.
Answer Authority	Non-authoritative answer	Response was retrieved from recursive cache within TTL, reducing resolver overhead.

Infrastructure Intelligence: Mapping the FQDN to `192.232.216.135` establishes the network focal point for all subsequent active fingerprinting steps, enabling reverse IP lookups, SSL certificate validation, and port-specific service probing.

3.3 Task 3: Web Application & Technology Stack Fingerprinting (`whatweb`)

Objective: Actively fingerprint the web application technologies, CMS framework, plugins, caching architecture, and disclosed administrative contact details.

```
$ whatweb networkwalks.com
```

```
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US],
HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/],
UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]


https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US],
Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client],
IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-
Protocol[website], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-
endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
```

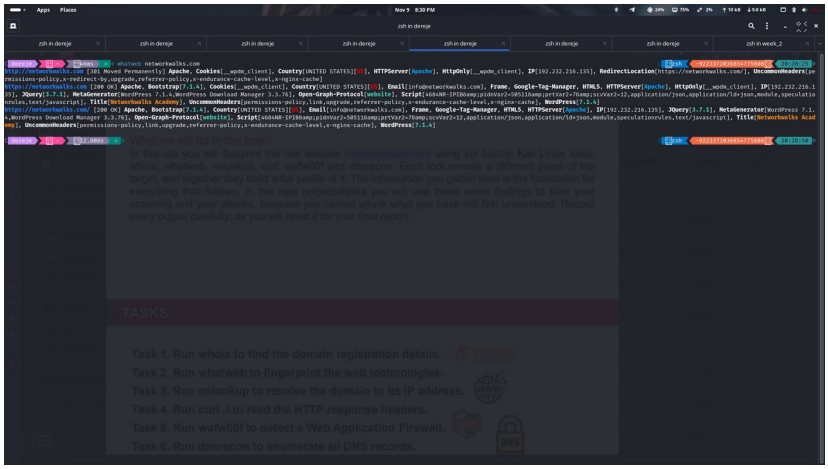


Figure 3.3: whatweb fingerprinting revealing WordPress 7.1.4, Apache, Nginx reverse cache, and plugins.

Identified Component	Disclosed Version / Value	Security Risk & Threat Implication
Content Management System	WordPress 7.1.4 (Core)	Meta generator exposure permits attackers to query CVE vulnerability databases for CMS core flaws.
Installed Plugin	WordPress Download Manager (v3.3.76)	Third-party plugins constitute the most frequent entry vector in WordPress web application breaches.
HTTP Web Server	Apache HTTP Server	Server banner exposed; detailed version string should be suppressed in production configuration.
Reverse Proxy & Cache	Nginx Reverse Cache + Endurance Cache	Indicates an Nginx edge proxy caching dynamic WordPress content generated by Apache backend.
Disclosed Contact Email	`info@networkwalks.com`	Provides a validated organizational target address for phishing and credential-stuffing campaigns.

3.4 Task 4: HTTP Protocol & Security Headers Analysis (`curl`)

Objective: Inspect raw HTTP response headers using `curl -I` to analyze redirection behavior, caching headers, session cookie security flags, and defensive header configurations.

```
$ curl -I networkwalks.com
```

```
HTTP/1.1 301 Moved Permanently
Date: Mon, 09 Nov 2026 17:36:04 GMT
Server: Apache
Permissions-Policy: private-state-token-redemption=(self ...), private-state-token-issuance=(self ...)
Expires: Mon, 09 Nov 2026 18:36:04 GMT
Cache-Control: max-age=3600
X-Redirect-By: WordPress - Really Simple Security
Set-Cookie: __wpdm_client=b66611d7fbd21602db8821fe9670c6b9; path=/; domain=networkwalks.com; HttpOnly
Upgrade: h2,h2c
Connection: Upgrade
Location: https://networkwalks.com/
Referrer-Policy: no-referrer-when-downgrade
X-Endurance-Cache-Level: 0
X-nginx-cache: WordPress
Content-Type: text/html; charset=UTF-8
```

Security Header / Parameter	Observed Configuration	Risk Evaluation & Security Impact
Strict-Transport-Security (HSTS)	Missing	Leaves initial plaintext HTTP connections open to SSL-stripping and man-in-the-middle attacks.
Content-Security-Policy (CSP)	Missing	No white-list policy for script sources, increasing vulnerability to Cross-Site Scripting (XSS).
X-Frame-Options / frame-ancestors	Missing	Site can be embedded in attacker-controlled iframes, creating clickjacking vulnerability.
X-Content-Type-Options	Missing	Lacks `nosniff`, leaving browsers vulnerable to MIME-confusion attacks.
X-Redirect-By Header	Disclosed	Explicitly discloses the active security plugin: `WordPress - Really Simple Security`.
Cookie Security (`__wpdm_client`)	Partial	`HttpOnly` is present; however, the `Secure` and `SameSite` flags are missing.

Header Remediation Priority: The web server correctly enforces a 301 redirection from HTTP to HTTPS. However, without an HSTS header (`Strict-Transport-Security: max-age=31536000; includeSubDomains; preload`), browsers will always initially attempt unencrypted HTTP connections, exposing user traffic to rogue Wi-Fi interception.

3.5 Task 5: Web Application Firewall Detection (`wafw00f`)

Objective: Fingerprint web perimeter defenses using `wafw00f` to identify active Web Application Firewall (WAF) solutions inspecting inbound HTTP/HTTPS traffic.

```
$ wafw00f networkwalks.com
```

```
                    ~ WAFW00F : v2.4.2 ~
        The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
```

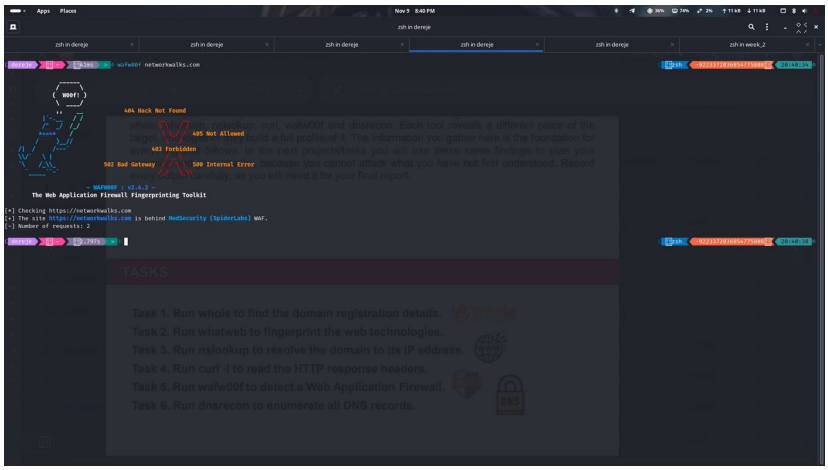


Figure 3.4: Wafw00f successfully identifying ModSecurity (SpiderLabs) protecting networkwalks.com.

WAF Feature	Observed Behavior	Penetration Testing & Defensive Impact
Detected WAF Engine	ModSecurity (SpiderLabs)	The industry standard open-source web application firewall module integrated into Apache/Nginx.
Fingerprint Accuracy	2 Probing Requests Sent	WAF detected with high confidence via behavioral blocking headers and HTTP error responses.
Defensive Capabilities	Real-time Payload Inspection	Filters OWASP Top 10 web vulnerabilities including SQL Injection, XSS, RFI/LFI, and command injection.
Exploitation Impact	Automated Scanners Blocked	Automated vulnerability scanners (Nikto, SQLmap) will be detected and blocked unless rate-limited and obfuscated.

WAF Defense Analysis: ModSecurity provides essential perimeter defense for the WordPress application. However, security administrators must ensure the OWASP Core Rule Set (CRS) is regularly updated to prevent bypasses utilizing HTTP parameter pollution or advanced character encoding.

3.6 Task 6: Advanced DNS Enumeration & Zone Analysis (`dnsrecon`)

Objective: Perform active DNS reconnaissance using `dnsrecon` to interrogate SOA, NS, MX, A, TXT, and SRV records, check DNSSEC implementation, and identify DNS daemon software versions.

```
$ dnsrecon -d networkwalks.com
```

```
2026-11-09T20:46:15 INFO Starting enumeration for domain: networkwalks.com
2026-11-09T20:46:15 ERROR No answer for DNSSEC query for networkwalks.com
2026-11-09T20:46:16 INFO         SOA ns6135.hostgator.com 50.87.144.87
2026-11-09T20:46:16 INFO         NS ns6135.hostgator.com 50.87.144.87 | Bind Version "9.16.23-RH"
2026-11-09T20:46:16 INFO         NS ns6136.hostgator.com 192.232.216.131
2026-11-09T20:46:20 INFO         MX mail.networkwalks.com 192.232.216.135
2026-11-09T20:46:21 INFO         A networkwalks.com 192.232.216.135
2026-11-09T20:46:21 INFO         TXT v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-11-09T20:46:25 INFO         SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net (443)
2026-11-09T20:46:26 INFO 8 Records Found & Completed enumeration for domain: networkwalks.com
```

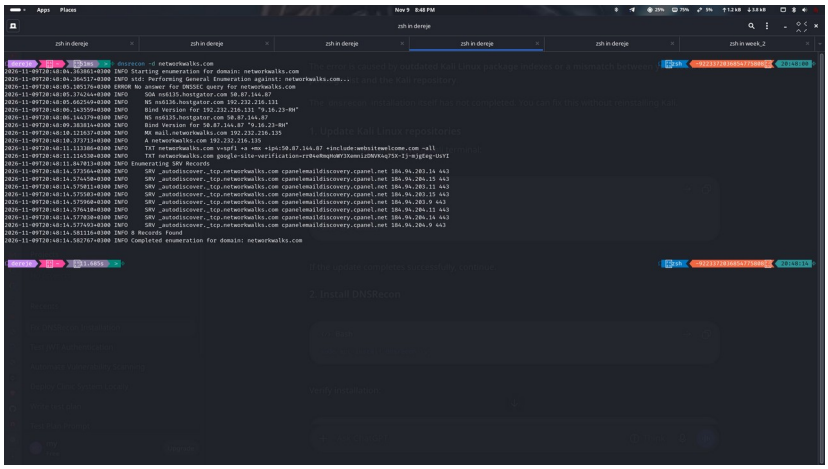


Figure 3.5: dnsrecon output enumerating DNS records, mail infrastructure, and BIND version banner.

Discovered Resource Record	Extracted Technical Value	Security & Operational Assessment
BIND Daemon Software Version	`"9.16.23-RH"`	Information Disclosure Exposes exact BIND build and Red Hat OS family, enabling targeted CVE queries.
Authoritative Name Servers (NS)	`ns6135.hostgator.com` (50.87.144.87) `ns6136.hostgator.com` (192.232.216.131)	Redundant authoritative DNS architecture hosted across distinct HostGator subnets.
Mail Exchanger (MX)	`mail.networkwalks.com` → 192.232.216.135	Co-located on the same physical/virtual server as the primary web application.
SPF TXT Record	`v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all`	Authorizes mail from primary A/MX and HostGator relay infrastructure. Configured with SoftFail (`~all`).
SRV Records (8 entries)	`_autodiscover._tcp` → `cpanelemaildiscovery.cpanel.net:443`	Discloses use of cPanel automated mail discovery services.
DNSSEC Implementation	No answer (Unsigned)	DNS responses lack cryptographic integrity validation, enabling possible DNS spoofing.

3.7 Task 7: Subnet Discovery, Port Scanning & OS Fingerprinting (`zenmap`)

Objective: Perform active network layer host discovery, TCP SYN stealth port scanning, service version detection, and remote OS fingerprinting across the `10.0.0.0/24` laboratory subnet using Zenmap GUI (Nmap 7.991).

```
$ nmap -sS -T4 -A -v 10.0.0.0/24 (executed via Zenmap GUI)
```

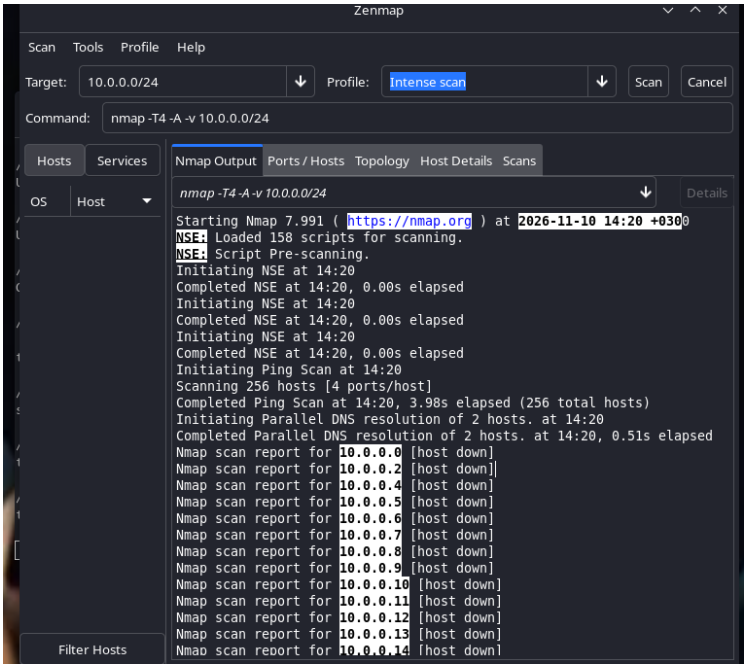


Figure 3.6: Zenmap GUI scanning 256 hosts on 10.0.0.0/24, detecting live hosts and OpenBSD 6.9 fingerprint.

Host IP Address	Status	Latency	Scanned Port Profile	OS & Network Detection Insights
10.0.0.1	Host Up	0.0010s	632 Closed (TCP RST) 368 Filtered (No-Response)	**OpenBSD 6.9** (CPE: `cpe:/o:openbsd:openbsd:6.9`). Traceroute: 2 hops away. Functions as internal security gateway.
10.0.0.3	Host Up	0.00096s	1000 Filtered (No-Response)	OS Fingerprint suppressed due to drop-all policy. Traceroute Hop 1: `10.175.10.1` (5.33ms), Hop 2: `10.0.0.3` (0.37ms).
10.0.0.0, 10.0.0.2, 10.0.0.4-255	Host Down	—	Unresponsive	254 IP addresses confirmed inactive or completely drop probes.

Network Security & Firewall Architecture Insights:

- **OpenBSD 6.9 Packet Filter (10.0.0.1):** The detection of OpenBSD 6.9 confirms that `10.0.0.1` operates as a security gateway using OpenBSD's native **Packet Filter (PF)**. The coexistence of closed ports (sending TCP RST) and filtered ports (silent drop) reflects intentional firewall rule sets.
- **Silent Drop Defense (10.0.0.3):** With 1000/1000 filtered ports, `10.0.0.3` enforces a strict drop-all policy for unsolicited inbound SYN packets, preventing remote OS detection and hiding potential internal services.

4. Consolidated Attack Surface Profile

Correlating the intelligence gathered across all seven reconnaissance tools allows us to assemble a comprehensive architectural model of the target organization's public and internal boundaries:

Target Perimeter	Asset & Parameter	Identified Specification	Security Status
External Web Boundary (networkwalks.com)	Public IP & Location	192.232.216.135 (Houston, TX, United States)	HostGator / Unified Layer ASN
	Web Server Daemon	Apache HTTP Server	Server banner exposed
	Content Management	WordPress 7.1.4 + Download Manager 3.3.76	Version meta generator exposed
	Perimeter Defense	ModSecurity (SpiderLabs) WAF	Active inspection & filtering
	Reverse Proxy / Cache	Nginx Reverse Cache + Endurance Cache	Disclosed in custom response headers
	Administrative Contact	`info@networkwalks.com`	Disclosed in HTML body
DNS Infrastructure (HostGator NS)	Name Servers	ns6135.hostgator.com / ns6136.hostgator.com	Redundant authoritative servers
	DNS Daemon Version	BIND 9.16.23-RH	CHAOS version query answered
	DNSSEC Integrity	Unsigned	Susceptible to DNS spoofing / cache poisoning
Mail Infrastructure	Mail Exchanger (MX)	mail.networkwalks.com (192.232.216.135)	Co-hosted on primary web host
	Sender Policy Framework	v=spf1 +a +mx +ip4:50.87.144.87 ~all	Configured with SoftFail (~all)
Internal Subnet (10.0.0.0/24)	Host 10.0.0.1	OpenBSD 6.9 Security Gateway	632 Closed, 368 Filtered Ports
	Host 10.0.0.3	Lab Host / Internal Appliance	1000 Filtered Ports (Silent Drop)

Perimeter Architecture Analysis: The external perimeter relies on a layered architecture: Cloudflare public resolution → HostGator BIND authoritative DNS → ModSecurity WAF inspection → Nginx reverse proxy caching → Apache backend web server → WordPress CMS core. While the WAF provides robust dynamic filtering, configuration-level information disclosure remains the primary risk factor.

5. Vulnerability Assessment & Risk Matrix

Based on the technical intelligence acquired during the engagement, each identified vulnerability has been evaluated, mapped to **Common Weakness Enumeration (CWE)** identifiers, and categorized by severity:

Finding ID	Vulnerability Description	CWE / Ref	Severity	Threat Vector & Potential Exploitation
SEC-01	BIND Software Version Disclosure (`9.16.23-RH`)	CWE-200	Medium	Adversaries query the CHAOS class to retrieve exact software versions, enabling targeted vulnerability research against known BIND CVEs.
SEC-02	Missing HTTP Security Headers (HSTS, CSP, XFO)	CWE-319 CWE-1021	Medium	Missing HSTS enables SSL-stripping MITM attacks. Missing CSP and X-Frame-Options leaves users vulnerable to XSS and clickjacking.
SEC-03	Absence of DNSSEC Cryptographic Signing	CWE-345	Medium	Without DNSSEC zone signing, responses can be spoofed or poisoned on intermediate DNS resolvers.
SEC-04	CMS & Plugin Version Information Leakage	CWE-200	Low	Disclosing exact versions of WordPress and Download Manager simplifies vulnerability scanning for automated threat actors.
SEC-05	Permissive SPF SoftFail Configuration (`~all`)	CWE-290	Low	SoftFail allows unauthorized senders to deliver spoofed emails into recipient spam folders rather than forcing a hard rejection (`-all`).
SEC-06	Co-Location of Mail and Web Services	CWE-657	Low	Web and mail daemons share public IP 192.232.216.135; service disruption or resource exhaustion on one affects the other.

Risk Summary: No critical remote code execution (RCE) vulnerabilities were uncovered during this reconnaissance phase, largely due to ModSecurity WAF enforcement. However, multiple medium-severity information disclosure and header configuration deficiencies provide actionable intelligence to attackers and should be remediated promptly.

6. Actionable Security Hardening Roadmap

To mitigate the risks identified during reconnaissance, the following prioritized security hardening measures are recommended:

1. **Suppress BIND Software Version Leakage:** Configure the BIND DNS server (`named.conf`) to reject or suppress version queries:

   ```
   options { version none; };
   ```

2. **Enforce Robust HTTP Defensive Headers:** Add security headers to the Apache/Nginx web configuration:

   ```
   Strict-Transport-Security: max-age=31536000; includeSubDomains; preload
   Content-Security-Policy: default-src 'self'; script-src 'self' 'unsafe-inline' https://www.google-analytics.com;
   X-Frame-Options: SAMEORIGIN
   X-Content-Type-Options: nosniff
   Referrer-Policy: strict-origin-when-cross-origin
   ```

3. **Implement DNSSEC Zone Signing:** Sign the DNS zone with Key Signing Keys (KSK) and Zone Signing Keys (ZSK), publishing the Delegation Signer (DS) record to GoDaddy.
4. **Harden SPF and Implement DMARC:** Upgrade the SPF record to HardFail (`-all`) and deploy a DMARC monitoring policy:

   ```
   v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com -all
   v=DMARC1; p=reject; sp=reject; pct=100; rua=mailto:dmarc@networkwalks.com;
   ```

5. **Suppress WordPress Generator Tags & Plugin Banners:** Remove version meta tags by adding a filter in the theme's `functions.php`:

   ```
   remove_action('wp_head', 'wp_generator');
   ```

Hardening Verification: Following implementation of the recommendations above, validation re-scans should be executed using `curl -I`, `dnsrecon`, and `whatweb` to verify that version banners are suppressed and defensive headers are actively returned.

7. Internship Reflection & Learning Outcomes

Completing the Week 2 project provided essential practical experience across offensive and defensive reconnaissance methodologies. By correlating evidence from multiple tools, I developed a deeper understanding of how modern organizations configure their public web perimeter, authoritative DNS, and private network segmentation.

Core Competencies Developed:

- **Passive OSINT Analysis:** Gained proficiency querying domain registries, analyzing EPP security flags, and mapping registrar privacy shields.
- **DNS Infrastructure Mapping:** Developed skills inspecting SOA, NS, MX, TXT (SPF), and SRV records to understand organizational mail flow and authentication standards.
- **Web Perimeter & WAF Fingerprinting:** Successfully detected ModSecurity (SpiderLabs) WAF filtering and identified missing HTTP security headers.
- **Network Scanning & Stack Fingerprinting:** Applied Zenmap/Nmap to survey the `10.0.0.0/24` subnet, analyzing TCP SYN responses and identifying OpenBSD 6.9 Packet Filter configurations.
- **Technical Reporting:** Learned to structure technical findings, correlate intelligence across data sources, and provide actionable security recommendations.

Intern Statement & Declaration:

This report represents original hands-on laboratory work completed for the **NetworkWalks Cybersecurity Internship Program (Batch B084)**. All assessments adhered strictly to ethical guidelines and authorized lab boundaries.

Submitted by: Dereje • Cohort B084 • November 2026